Tommy Lin

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Education

National Yang Ming Chiao Tung University (NYCU), Taiwan Sept 2023 – Jun 2027 (expected)
B.S. in Computer Science | Cumulative GPA 4.13 / 4.3

- **Relevant Coursework:** Introduction to AI (path search, a Connect-4 agent, diffusion image generation, multi-armed bandits), Embedded Systems (Raspberry Pi), Digital Circuit Design (FPGA), Signals & Systems, Differential Equations.

National University of Singapore (NUS) — Semester Exchange Aug 2026 – Dec 2026
Exchange student, Semester 1 AY2026/27 (currently at NUS).

Projects

SAFMC 2026 — High-Speed Drone Flock 2026

- Designed to drive a flock of 2–3 drones through a gate course autonomously — an FSM + Mediator schedules each drone and keeps the flock collision-free.
- As **Project Lead**, directed full-system evaluation and design (avionics, software stack, strategy); validated the full autonomy stack in Gazebo simulation, and built and flight-tested the two-drone platform.

SAFMC 2025 — Category D2 2025

- Four autonomous drones that search an unknown indoor space — localizing, avoiding obstacles, and finding hidden rooms — to deliver supplies in a search-and-rescue mission.
- As a **team member**, contributed to the ROS 2 / PX4 stack, BUG / APF path planners, ArUco-marker detection, and Gazebo simulation; team placed **4th**.

ExpressLRS Long-Range RC Radio Modules 2026

- ExpressLRS is an open-source, low-latency long-range radio-control link for drones and RC aircraft.
- Self-designed the transmitter (TX) and receiver (RX) modules (STM32 + SX1280) for this link, through full KiCad PCB layout; currently in pre-validation.

Technical Skills

Robotics & Vision: ROS 2, PX4, Gazebo, OpenCV
Hardware & EE: PCB design (KiCad), embedded systems, electronic system evaluation & design, manufacturing & quality testing

Awards

SAFMC 2025 — 4th Place, Singapore Amazing Flying Machine Competition
CTCI Foundation Scholarship — Creativity Award (2025)